

# Presentation prep — read this before you go up

Score: 59.7 / 100. D1 12.1/25 · D2 29.8/35 (1st of 28) · D3 14.2+/40 Started the day at 0. Was 49.9 an hour before the end.

## 1. What we built — say this first

A windowed vision-language pipeline that watches long video and reports anomalies with timestamps — plus a frozen-encoder classifier that catches what the VLM is blind to.



The one-sentence version of why it works: The VLM finds *appearance* — fire, smoke, flood, crashes. The frozen encoder finds *duration* — loitering, stalled vehicles. Neither does both.

## 2. The three-agent system — how we actually worked

Three Claude Code sessions, on **one repository**, talking to each other by direct message.

| Agent                    | Role                                                              | Owned                             |
|--------------------------|-------------------------------------------------------------------|-----------------------------------|
| Agent 1<br>(coordinator) | Research, strategy, all GPU runs, all commits                     | src/vad/ , modal_app.py , status/ |
| Agent 2 (builder)        | Data pipeline, synthetic video generation, training, scoring code | src/ahc_vad/ , scripts/ , tests/  |
| Agent 3 (comms)          | Demo, slides, process doc                                         | demo/ , slides/ , docs/           |

The rule that made it work: **ownership by directory**. No agent edits another's files. Cross-boundary changes are *requested*, not made.

We learned that the hard way. Early on, Agent 1 and Agent 2 both built the same four modules — scoring, submission, event types. We threw away Agent 1's copies. After that, one rule fixed it.

**One committer.** Agent 1 reviewed and pushed everything. The repo is a graded deliverable; three agents committing in parallel would have made the history unreadable.

**They corrected each other, out loud.** Real examples worth quoting: - Agent 2 caught Agent 1 using **first-fit instead of best-loU** matching in the scorer — silently wrong whenever predictions overlap. - Agent 3 caught a **Gemini run that exited "successfully"** having done 6 of 34 videos, writing the 28 failures to disk as confident "normal" predictions. - Agent 1 published *"bigger is worse, 4B beats 8B"*, then

**retracted it** when asked for the mechanism — the models differ on 3 videos out of 24,  $p = 0.25$ . **That retraction was load-bearing: the 8B produced our best score.**

### 3. The knowledge factory — how three agents stayed coherent

Nothing important lived in chat. Everything went into files.

| File                               | What it is                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>.context/</code>             | <b>Shared memory.</b> Problem, dataset, prior art, decisions. Every finding written down, with provenance: <i>measured</i> , <i>read in a paper</i> , or <i>assumed</i> .                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>status/ledger.md</code>      | <b>The experiment ledger.</b> Every technique across 7 tiers — input, prompting, decoding, frozen heads, fine-tuning, RL, cascades — marked  <b>done</b> /  <b>running</b> /  <b>planned</b> /  <b>rejected with the reason.</b> |
| <code>status/models.md</code>      | Why each model was chosen, what it was compared against, what we rejected before testing.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <code>status/experiments.md</code> | Every run and its verdict, including the failures.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <code>plans/</code>                | Specs written before code.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

**Why it mattered:** an agent could crash, restart, and pick up from the files. That happened. Twice.

### 4. Models we tested — the headline table

All five on an identical 24-video set, same prompt, same policy, Modal A100.

| Model                  | Size   | Precision | Recall | F1   | Verdict                                                                                                 |
|------------------------|--------|-----------|--------|------|---------------------------------------------------------------------------------------------------------|
| Qwen3-VL-8B-Instruct   | 8B     | 0.62      | 0.40   | 0.48 | <b>Shipped — produced our D1 + D2</b>                                                                   |
| Qwen3-VL-4B-Instruct   | 4B     | 0.77      | 0.50   | 0.61 | Ties the 8B (differ on 3 of 24)                                                                         |
| Cosmos-Reason2-8B      | 8B     | 0.57      | 0.20   | 0.30 |  Half the recall    |
| Qwen3-VL-2B-Instruct   | 2B     | 0.30      | 0.15   | 0.20 |  Too small          |
| Qwen2.5-VL-7B-Instruct | 7B     | 0.00      | 0.00   | 0.00 |  Silent on 20 of 24 |
| SigLIP2 + GRU head     | frozen | —         | —      | —    | <b>Shipped — +2.4 marks</b>                                                                             |

Three findings worth saying out loud:

- Benchmark leadership does not transfer.** Cosmos-Reason2-8B **tops NVIDIA's own Traffic Anomaly Reasoning leaderboard** and got *half the recall of a 4B* on our task.
- Qwen2.5-VL-7B is ASK-HINT's own published backbone** (89.83% AUC on UCF-Crime in the paper). Here it returned a bare `[]` on 20 of 24 videos. Instruction-following failure, not blindness.
- We nearly didn't test SigLIP.** It was marked  *rejected* in our ledger on the theory that a frozen probe can't use context. Four of the top five leaderboard entries were running it. **The board was evidence; our theory was wrong.**

**Also tried and rejected before testing:** hosted models (banned from the runtime path), 72B/90B (too large), Llama-3.2-Vision (accepts **one image per request**; we send 8–32 frames).

---

## 5. Platforms

| Platform          | Used for                                | Outcome                                                               |
|-------------------|-----------------------------------------|-----------------------------------------------------------------------|
| Modal             | Every GPU job — A100-40GB and A100-80GB | ✅ The workhorse. ~\$12 of \$30 credit.                                |
| Google Gemini API | Reference ceiling, dev-time only        | ⚠️ Works, but no trustworthy number — two runs killed by our own bugs |
| NVIDIA NIM        | Hosted VLM catalogue                    | ❌ 4 of 5 models 404 — account entitlement, takes days                 |
| HuggingFace       | Model weights, incl. gated Cosmos       | ✅                                                                     |
| Kaggle            | Free T4 fallback                        | Authenticated, never needed                                           |
| Lightning AI      | Considered                              | Not used — Modal covered it                                           |

**Why Modal:** one Python file, `@app.function(gpu="A100")`, jobs run headless and detached. Persistent volumes held the dataset so nothing re-uploaded. **I could drive it entirely from the terminal**, which is what made three agents on one GPU budget workable.

---

---

## 6. The approach — and the two things that actually moved the score

**+9.8 marks came in the final hour. Fourteen VLM interventions before that moved nothing.**

### The big one: span geometry, +6.1

Eight prediction sets — three model sizes, five window configs, a fine-tune, targeted prompts — **all scored exactly 8.0 on D3.**

That invariance was the clue. **If changing the model changes nothing, the failure isn't in the model.**

Our D3 spans were **1–8 seconds inside videos of 327–602 seconds**. A 5-second span can never reach 50% overlap with a 100-second event — *geometrically impossible, no matter how right the class is*.

We widened spans to 112–138s. **49.9 → 56.0**. No new inference. No changed class. **Our classes had been right all along.**

The evidence was in our own notes from that morning: the practice data had a **125-second** congestion event. We wrote down that D3 events are long — then spent hours swapping models instead of spans.

### The second: SigLIP2, +2.4

Frozen SigLIP2 → 8 frames → bidirectional GRU head → 12 classes. It predicts **loitering** on two videos — a class we score **0/6** on and **no VLM proposed all day**.

---

---

## 7. What we'd say about the failures

**Ten silent failures in one day. None crashed.** Every one produced a plausible artifact.

Pick two or three of these to tell: - **ffmpeg's -ss is a keyframe seek** — silently returns *more* than you ask for. A 468s video rendered as 241s and every timestamp after it was wrong. - **The fix for one bug**

**caused another.** We added per-video error handling so one bad video couldn't kill a 34-video run. It then converted a loud crash into **10 rows of confident, plausible, wrong output.** - **SigLIP reported "validation accuracy 1.000".** Meaningless — 77 embeddings and a 7-sample validation split, because a 13GB upload had died unnoticed. - **We deleted 125 synthetic videos we had already rendered** because their timestamps were silently wrong. *We threw away a third of our training data rather than train on labels we couldn't verify.*

**The line to land:**

**In a seven-hour build, every failure that mattered was silent. So we validated intermediate artifacts, not outputs.**

---

## 8. Likely questions

**"Why not just fine-tune?"** We did. LoRA on Qwen3-VL-4B, 300 steps, loss converged. It scored **42.4** replacing the baseline and **49.2** added to it — both **below** the 49.9 zero-shot. It trained on 2,610 rows instead of the full set because an upload failed, and 71% of its targets were empty, teaching it to stay silent when our problem was recall.

**"Is it real-time?"** Measured, not asserted. **A window closes every 10 seconds and is scored in 1.83 seconds — 5.5x headroom, 2.8x at p95.** Self-reported latency, instrumented from the first call.

**"What would you do next?"** 1. **GRPO with a tIoU-shaped reward** — the literature says it beats SFT on temporal grounding, and 75 of the 100 marks are IoU-gated. 2. **Make SigLIP the cheap always-on stage** and the VLM the verifier. That's the cascade the brief actually asks for, and it's what the leader is running. 3. **Feed SigLIP the full dataset** — it earned +2.4 marks trained on a fifth of the data.

**"What are you still bad at?"** Four classes we never detect: `loitering` 0/6, `fighting` 0/3, `road_spill` 0/3, `stalled` 0/2. All **context and duration** classes. The VLM is confident on appearance and blind to anything defined by time rather than pixels.